

SHAFAYET KHAN SHAFEE

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☎ +880-1791-051104 📍 Dhaka, Bangladesh

Data Scientist with 3 years of experience using experimentation, product analytics, and causal inference to drive product and growth decisions in fintech. Experienced in partnering with product, engineering, and business stakeholders to design experiments, analyze user behavior, build analytics infrastructure, and translate complex analyses into actionable recommendations. Strong SQL, Python, and statistical modeling skills with experience owning end-to-end analytical projects in production.

Professional Experience

Data Scientist II

Pathao Pay

Dhaka, Bangladesh

Jan 2025 – Present

- Designed and executed end-to-end randomized experiments across fintech product and marketing initiatives, spanning both user and merchant populations. Translated experiment findings into product and growth decisions by identifying actionable interventions, detecting ineffective interventions early, and preventing unnecessary promotional spend.
- Built and maintained a centralized analytics transformation layer from raw source data using dbt and BigQuery on GCP, designing reusable staging and aggregated data models to make source data analysis-ready. Standardized metric definitions and transformation logic across Growth, Product, Operations, and Finance teams, reducing repeated analytical work and improving consistency of business reporting.
- Evaluated the causal impact of a fintech product rollout on merchant business outcomes in a non-randomized setting using quasi-experimental methods, including propensity score matching, inverse probability weighting, and Difference-in-Differences.
- Improved controlled rollout planning by applying Bayesian experimental design to determine sample sizes using probabilistic criteria, making tradeoffs between statistical confidence, operational cost, and experiment efficiency explicit and communicable to stakeholders.
- Evaluated product adoption across small and unequal-sized cohorts using Bayesian modeling, enabling more reliable estimates under limited data by quantifying uncertainty and supporting risk-aware rollout decisions.

Graduate Data Scientist

Pathao Limited

Dhaka, Bangladesh

Jul 2023 – Dec 2024

- Designed and executed end-to-end randomized experiments to optimize BNPL repayment engagement, testing notification strategies across thousands of users. Demonstrated the impact of increased repayment communications in reducing due unpaid user rate, contributing to a 4% reduction after broader rollout without negatively impacting monthly active users.
- Built automated reporting workflows using dbt, BigQuery, Looker Studio, and Google Apps Script, delivering executive dashboards used by Fintech and Finance leadership to monitor key business metrics. Reduced manual reporting overhead and improved access to timely business insights.
- Developed a causal ML-based personalization framework to estimate heterogeneous treatment effects across user subgroups. Enabled targeted promotion strategies by identifying high-response segments, contributing to a 25% reduction in monthly promotional spend while maintaining comparable user activity and business performance.
- Co-developed a repayment collection prioritization framework to estimate time-to-repayment and stratify overdue BNPL users by repayment likelihood. Optimized limited CX call capacity by prioritizing users with higher likelihood of repayment following outreach while reducing unnecessary collection efforts.
- Co-developed a credit risk scoring framework for BNPL applicants to estimate probability of default. Enabled risk-based user segmentation and dynamic credit limit assignment for 100K+ users, allowing the business to expand BNPL access while managing credit risk exposure.

Technical Skills

- **Programming Languages:** Python, R, SQL.
- **Machine Learning:** Ensemble Methods (XGBoost, CatBoost, LightGBM, BART), Survival Modeling, Conformal Prediction, Model Calibration, Causal ML (DML, Causal Forest, BCF).
- **Statistics:** Hypothesis Testing, Bayesian Modeling, Causal Inference (IPW, Matching, DiD), A/B Testing, Time Series Analysis.
- **Data & ML Engineering:** dbt, Kedro, MLflow, Bash Scripting.
- **Tools & Platforms:** BigQuery, GCP, Docker, Git, CI/CD (GitHub Actions, GitLab CI).
- **Analytics & Visualization:** Looker Studio, MixPanel, R-Shiny.

Education

M.Sc. in Applied Statistics GPA 3.97 / 4.00 ISRT, University of Dhaka	Dhaka, Bangladesh 2022–2023
B.Sc. in Applied Statistics CGPA 3.96 / 4.00 ISRT, University of Dhaka	Dhaka, Bangladesh 2018–2022

Publications

[1] **S. K. Shafee** and M. S. Rahman. *On the estimation of the median odds ratio for measuring contextual effects in multilevel binary data from complex survey designs*. [Under review, 2026]. arXiv: [2606.15145 \[stat.ME\]](#).

[2] **S. K. Shafee** et al. *G-computation for causal effect estimation from observational hierarchical data with unmeasured cluster context*. [Under review, 2026]. arXiv: [2606.14131 \[stat.ME\]](#).

[3] M. N. I. Sium et al. *Quantifying Inconsistent Mediation: The Mediated Magnitude Share and the Degree of Suppression*. [Under review, 2026]. arXiv: [2609.33846 \[stat.ME\]](#).

[4] **S. K. Shafee** et al. “Investigating the causal effect of maternal continuum of care on child’s minimum acceptable diet: A multilevel approach using partially pooled propensity score weighting”. In: *PLOS ONE* 20.11 (2025), pp. 1–13. DOI: [10.1371/journal.pone.0335972](#).

Open Source

- [skmiscpy](#) — Python package for causal inference diagnostics, including Love plots, mirror histograms, and SMD computation for IPW workflows.
- [randomizer](#) — Dockerized Shiny app for experimental unit randomization with SRM testing and covariate balance diagnostics.
- [kedrogen](#) — CLI tool for scaffolding reproducible data science projects from cookiecutter templates.
- [python-uv-gcloud](#) — Minimal Docker base image with Python, uv, and Google Cloud SDK for CI/CD pipelines and cloud workflows.
- [MOR](#) — R package implementing post-estimation functions to compute the Median Odds Ratio and confidence intervals from fitted multilevel binary logistic models.

Awards & Achievements

- **Dean’s Award**, Faculty of Science, University of Dhaka 2025
- **Conference Award for Scientists**, 45th Annual Conference of the ISCB, Thessaloniki, Greece 2024
- **Datathon Finalist**, Top 10 of 300 teams; built a purchase-behavior prediction model for mobile data packages, Robi Axiata Limited, Dhaka, Bangladesh. 2024
- **National Science and Technology (NST) Fellowship**, M.Sc. thesis research in multilevel modeling, Ministry of Science and Technology (MoST), Government of Bangladesh 2023